

Evaluating PRIMA through OptiProfiler

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1 Introduction

This report tests the performance and correctness of PRIMA, a package for solving general nonlinear optimization problems without using derivatives.[2]

All experiments were performed under Linux (x86_64) with MATLAB R2025b. PRIMA[2] and OptiProfiler[1] were downloaded from their GitHub repositories and installed following the official instructions.

2 Rosenbrock Function Testing

After we make sure PRIMA itself can run correctly, we begin with the Rosenbrock Function with N variables, which has the form like

$$f(\mathbf{x}) = \sum_{i=1}^{N-1} \left[100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2 \right].$$

The initial point was set to $\mathbf{x}_0 = (-1, -1, \dots, -1)^\top \in \mathbb{R}^{20}$, and four constraint scenarios were considered. PRIMA was invoked with default options (double precision, classical mode). The results are summarized in Table 1.

Table 1: Results of PRIMA on the 5-dimensional Rosenbrock function under four constraint scenarios.

Constraints	Optimal $f(\mathbf{x})$
Unconstrained	0.00000000
$x_i \leq 0$	19.00000000
$\sum x_i \leq 1, x_i \geq 0$	17.41525838
$\sum x_i^2 \leq 1, x_i \geq 0$	16.32725289

In the unconstrained case, PRIMA successfully located the global minimum with $f = 0$, confirming that the solver can recover the known analytical optimum. Under the bound constraints $x_i \leq 0$, the search is forced toward the origin; the returned optimal value $f = 19 = N - 1$ agrees with the theoretical value of the Rosenbrock function at the zero vector, indicating that all constraints were satisfied. When linear or nonlinear constraints are imposed, the global minimizer $(1, \dots, 1)$ lies outside the feasible region, and the optimal values increase to approximately 17.4 and 16.3, respectively. This trend is consistent with the tightening of the feasible set. PRIMA automatically selected the appropriate backend solver and also revised the initial point to improve feasibility when necessary, demonstrating robust preprocessing.

3 OptiProfiler Test

To evaluate the impact of floating-point precision on PRIMA’s performance, two sets of benchmark tests were conducted using the OptiProfiler framework with the S2MPJ problem library. The first test compared the double-precision case against the single-precision one, and the second compared the double-precision case against the quadruple-precision one. Both tests used the same problem set: `pctype = 'ubln'`, `mindim = 2`, `maxdim = 10`. The tests were performed under `plain` and `noisy` characteristics separately. We lowered the maximum dimension because of the limited memory on a personal laptop. A total of 474 problems were selected per case; for the tests under the noisy feature, each problem was run five times per case.

The solver scores produced by OptiProfiler are reported in Table 2 and Table 3. Following the convention of performance profiles, a higher normalized score indicates better overall convergence performance within a solver pair.

Table 2: Relative scores for double and single precision.

Feature	Double precision	Single precision
plain	1.0000	0.8377
noisy	1.0000	0.9275

Table 3: Relative scores for double and quadruple precision.

Feature	Double precision	Quadruple precision
plain	1.0000	1.0000
noisy	1.0000	1.0000

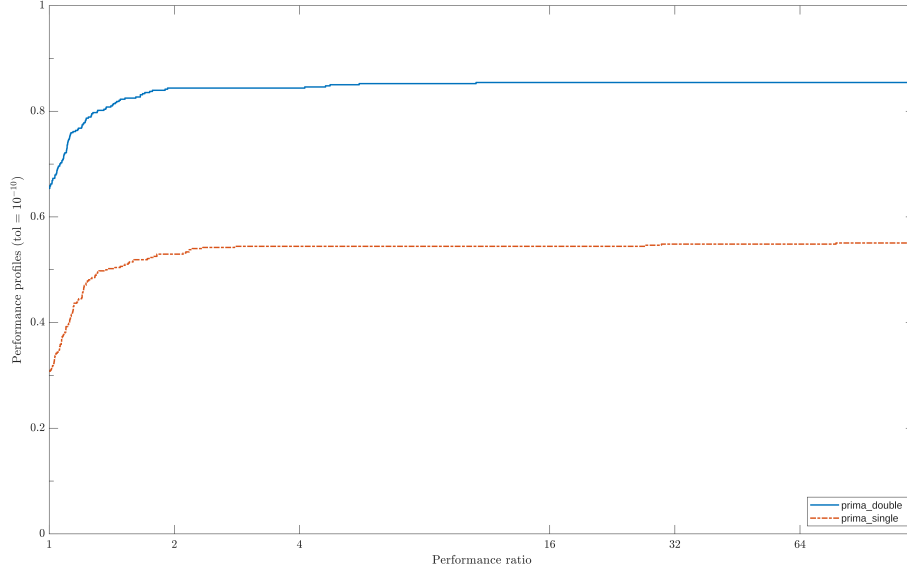


Figure 1: Performance profiles for double and single precision on the plain problem set.

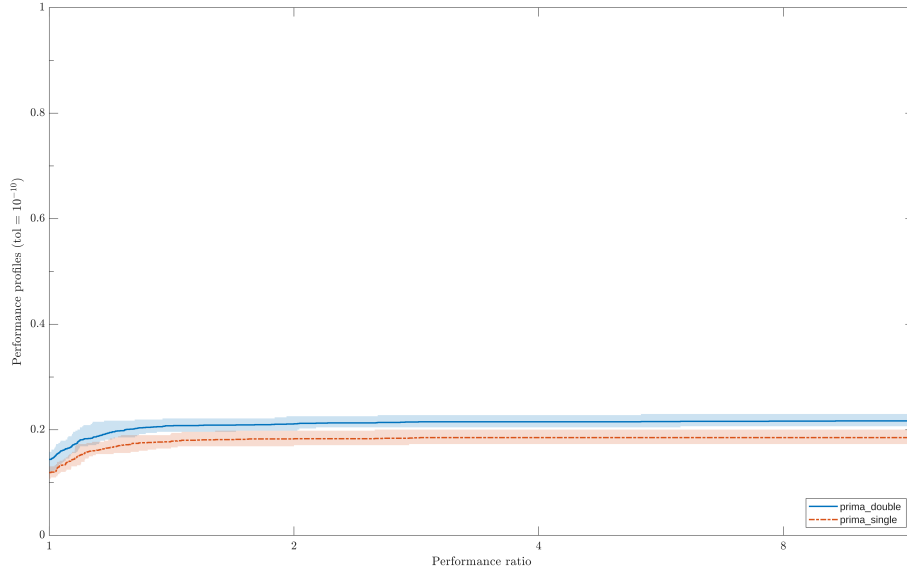


Figure 2: Performance profiles for double and single precision on the noisy problem set.



Figure 3: Performance profiles for double and quadruple precision on the plain problem set.



Figure 4: Performance profiles for double and quadruple precision on the noisy problem set.

Figure 1 and Table 2 compare the double-precision and single-precision cases on the plain and noisy problem sets. In both settings, double precision achieves a higher score and a superior performance profile, with the advantage being particularly evident in the

noise-free case. When Gaussian noise is introduced, the gap between the two solvers narrows, but double precision remains the better-performing option.

In contrast, the double-precision and quadruple-precision cases show no meaningful difference. Their scores are identical, and their performance profiles overlap almost completely across all tolerances. This holds for both the plain and the noisy features, suggesting that double precision already provides sufficient numerical resolution for the tested problems.

All profiles shown in this section correspond to the strictest tolerance (10^{-10}). This tolerance was chosen because it magnifies the differences between solvers that would otherwise appear indistinguishable under looser convergence criteria. We can always notice that each precision degrades significantly in the noisy setting.

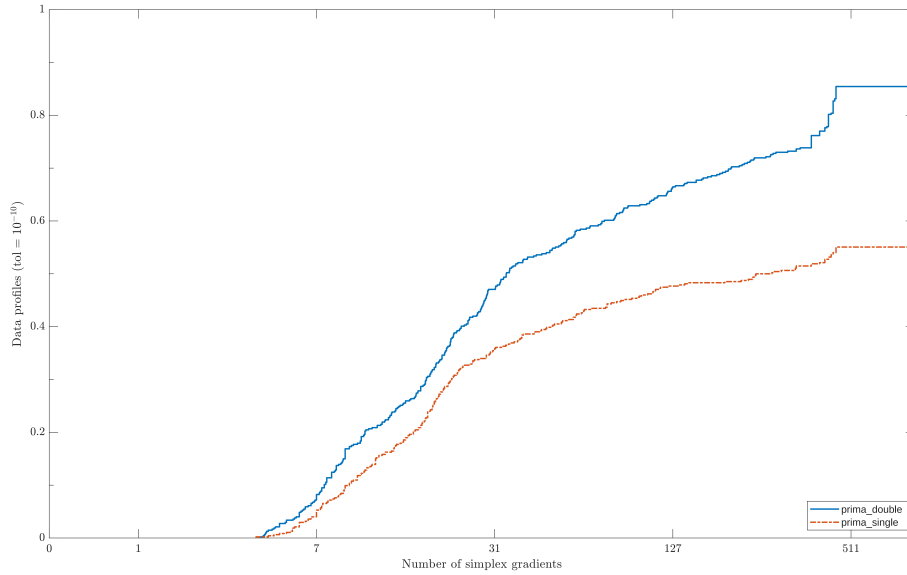


Figure 5: Data profiles for double and single precision on the plain problem set at tolerance 10^{-10} .

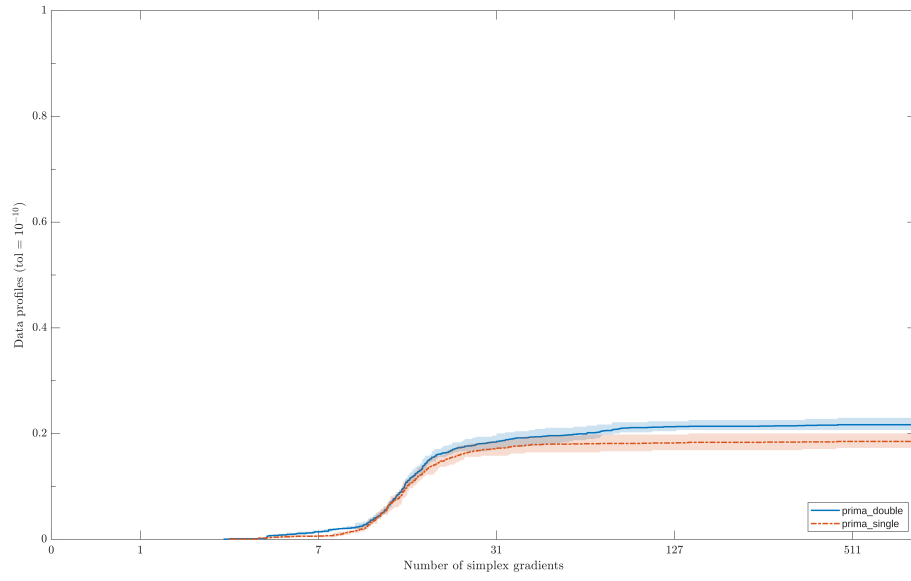


Figure 6: Data profiles for double and single precision on the noisy problem set.

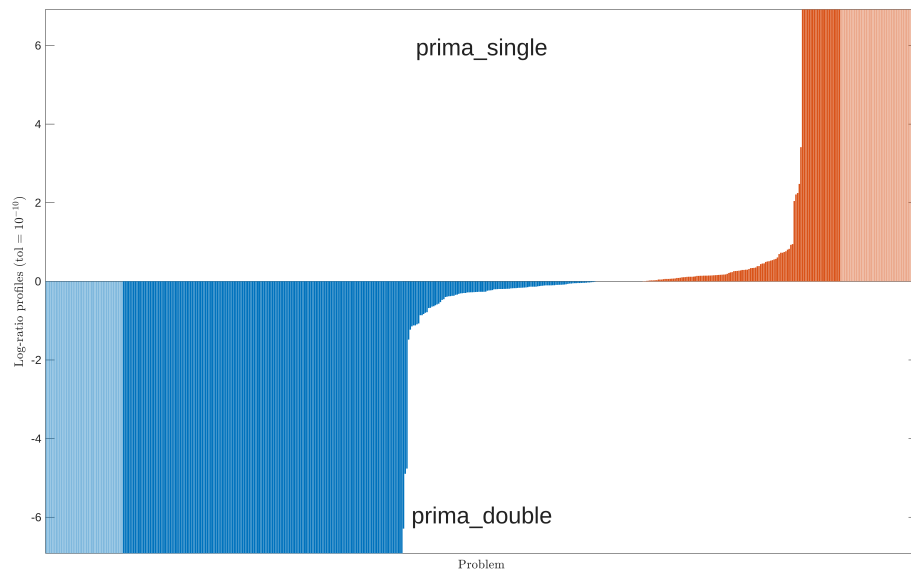


Figure 7: Log-ratio profiles for double and single precision on the plain problem set.

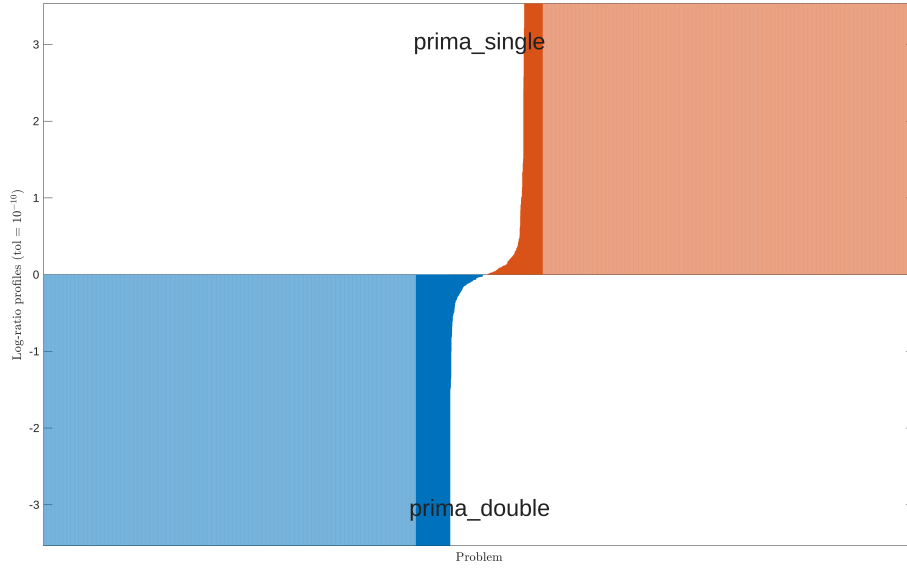


Figure 8: Log-ratio profiles for double and single precision on the noisy problem set.

The log-ratio profiles reinforce this view. Whether examined problem by problem or through aggregate scores, the conclusions remain the same. More problems favor the double-precision case over the single-precision one, especially at strict tolerances, while the gap shrinks under noise. Between the double-precision and quadruple-precision cases, the differences remain negligible throughout.

References

- [1] C. Huang, T. M. Ragonneau, and Z. Zhang. OptiProfiler: a platform for benchmarking optimization solvers. Unpublished, 2026.
- [2] Z. Zhang. PRIMA: Reference Implementation for Powell’s Methods with Modernization and Amelioration. available at <http://www.libprima.net>, DOI: 10.5281/zenodo.8052654, 2023.